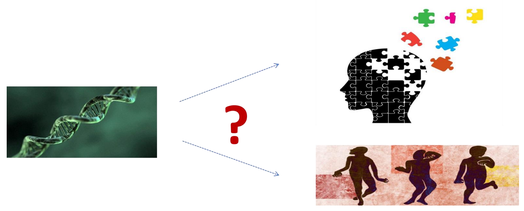
Estimating Disease Onset from Change Points of Markers Measured with Error

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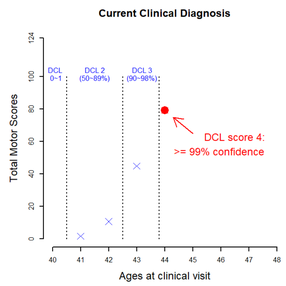
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Motivation: Huntington Disease (HD)



When is HD motor-onset ?

Two Scoring System for Clinical Diagnosis



- Total Motor Scores
- Diagnostic Confidence Level

Standard Clinical Onset of HD: A patient has experienced the onset of HD at age 44.

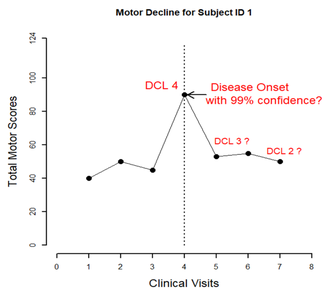
Challenges on Clinical Diagnosis



- Subjective Scoring System
  - Among-clinicians variability
  - Patients' comorbid conditions

Errors in "when HD motor onset occurs"

Example.



Nonlinear Location Shift-Marker Model and Our Goal

- Y_{ij} : total motor scores for subject i at jth clinical visit.
- s_{ij}: subject's age at clinical visit j.
- W_i is a p-dim'l subject specific baseline covariates.
- T_i: an inflection point of each subject's longitudinal trajectory, assumed to be unknown.
- Transform all times : $A_{ij} = log(s_{ij}),\ Z_i = log(T_i)$.

- *Model:*

$$Y_{ij} = \omega(A_{ij} - Z_i) + \varepsilon_{ij},\ \ \varepsilon_{ij} \sim normal(0, \sigma_\varepsilon^2),$$

$$Z_i = \beta^T W_i + u_i,\quad u_i \sim normal(0, \sigma_u^2).$$

- *Region dependent shape constraints on $\omega(\cdot)$:*

$$\omega''(A_{ij} - Z_i) > 0 \text{ for } A_{ij} < Z_i,\ \ \omega''(A_{ij} - Z_i) \le 0 \text{ for } A_{ij} \ge Z_i.$$

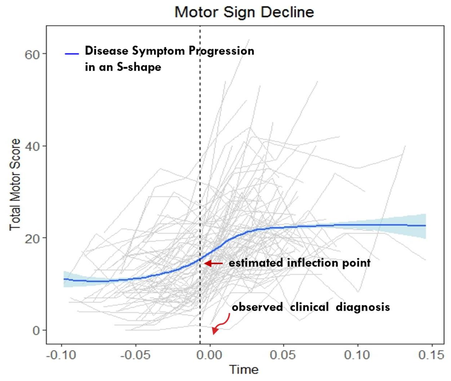
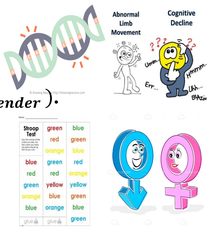


Fig1. Observed Longitudinal trajectories of total motor scores (gray curves) after all time points are shifted by their observed clinical diagnosis. Our goal is to estimate the longitudinal disease symptom curve and the inflection point as the objective measure of disease onset.

Application to the Huntington Disease (PREDICT-HD) Study

- Large observation study for 13 years (2001-2013).
- Assessed HD impairments from 1224 patients in 32 word-wide sites.
- Focus on 80 patients: genetically at-risk for HD and diagnosed after first visit. made at least 5 and at most 7 clinical visits.
- Standardized baseline covariates: $W_i = (1, W_{i,CAG}, W_{i,STMD}, W_{i,color}, W_{i,word}, W_{i,inter}, W_{i,gender})$.



- *Consider a Logistic Model for $\omega(\cdot)$:*

$$\omega(A_{ij}, W_i; \theta, u_i) = \theta_1/[1+\exp\{-\theta_2(A_{ij} - \beta^T W_i - u_i)\}]$$

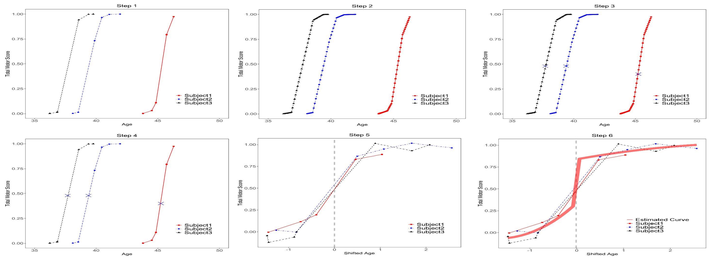
Inflection Point model Z_i

Fixed effects $\theta = (\theta_1, \theta_2, \beta^T)^T$ with

β: (p+1)-dim'l coef. vector

Random Effects $u_i \sim Normal(0, \sigma_u^2)$

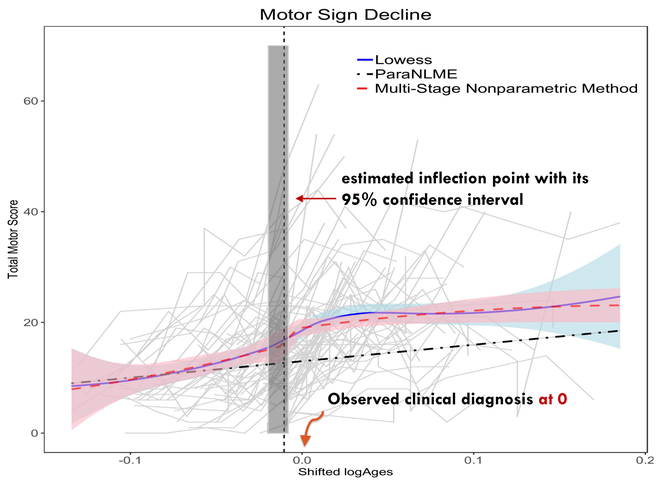
- Multi-Stage Nonparametric Method: iterative algorithm to estimate $\omega(\cdot)$ using shape constrained P-splines (Pya and Wood, 2015) and Z_i using the bisection method (Liao and Meyer, 2017).



Illustrate the multi-stage nonparametric procedure. At early step, we use piecewise linear model to compute individual motor sign decline curves and their inflection points. To do that, we generate $m* = 45$ interpolated time points. We use observed data only thereafter. We shift the observed time points by each subject's estimated individual inflection point and all subjects have the same inflection point at 0 (Step 5) and estimate the average of the longitudinal trajectories (red solid curve) and its infection point (dashed vertical line) based on all subjects' longitudinal observed data points. In this procedure, we use the ordinary least squares method for parameters β .

- Parametric NLME Method: estimate Z_i using best linear unbiased predictors.

Results of the PREDICT-HD Study



- Multi-Stage Nonparametric captures an "S"-shape, but parametric NLME cannot (θ_2=1.068).
- The estimated inflection points were more likely to occur before (0.5 year earlier) the clinical diagnosis.
- Length of CAG repeats and gender are correlated with estimated inflection points.

	Multi-Stage Nonparametric			Parametric NLME		
Param	Estimate	Std. Error	P-value	Estimate	Std.Error	P-value
β_0	3.81	0.020	0.000	4.32	0.904	0.000
β_{CAG}	**-0.16**	0.017	0.000	**-0.35**	0.072	0.000
β_{SDMT}	-0.01	0.021	0.620	0.04	0.057	0.506
β_{color}	0.00	0.026	0.921	0.14	0.078	0.083
β_{word}	0.00	0.025	0.729	0.08	0.068	0.222
β_{inter}	-0.06	0.021	0.059	0.02	0.057	0.687
β_{gender}	**0.09**	0.035	0.010	-0.03	0.096	0.795

Conclusion

Our method would provide a new measure of HD motor-onset.

- Detect ages at motor disease onsets earlier than ages at clinical diagnosis.
- Early motor diagnosis is important;
  - Clinicians: provide early treatment guidance to patients.
  - Patients: facilitate important life choices.

Selected References

- Pya, N. and Wood, S.N. (2015). Shape constrained additive models. *Statistics and Computing* 25, 543-559
- Liao, X. and Meyer, M.C. (2017). Change-point estimation using shape-restricted regression splines. *Journal of Statistical Planning and Inference* 188, 8-21.
- Paulsen, J.S. and others. (2014). Prediction of manifest Huntington's disease with clinical and imaging measures: a 12-year prospective observational study. *The Lancet Neurology*13, 1193–1201.

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